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Comparative Analysis of Gaussian Naive Bayes and Hybrid Models for Enhanced Classification

Kristina Bhandari¹, Namita Bhatta²

¹Institute Of Engineering, Thapathali Campus, Bagmati 446000 Nepal (e-mail: christinabhandari60@gmail.com)

Corresponding author: Kristina Bhandari(e-mail:christinabhandari60@gmail.com).

ABSTRACT In this study, we applied Naïve Bayes classification from scratch for Estimation of Obesity Levels Based on Eating Habits and Physical Condition dataset that contained both Categorical and Numeric data. Naïve Bayes Classifier is a probabilistic algorithm for classification based on Bayes theorem which assumes that the features are independent. We first used Gaussian Naïve Bayes Classification that achieved an accuracy of 0.64 for classification and Categorical Naïve Bayes Classification that achieved an accuracy of 0.71. Then to increase its accuracy, we explored various Hybrid models and compared its accuracy. We used a model that is a hybrid of Gaussian and Categorical Naïve Bayes Classifier. We used Gaussian Naïve Bayes for continuous data and Categorical Naïve Bayes for categorical data. This hybrid model significantly improved the classification performance with an accuracy of 0.99 for the dataset.

INDEX TERMS Naïve Bayes Classifier, Gaussian Naïve Bayes Classifier, Categorical Naïve Bayes Classifier

I. INTRODUCTION

The Naive Bayes algorithm is a simple yet powerful classification method based on Bayes Theorem of Probability that may be used to predict class membership for a given dataset. This algorithm assumes that all the features are independent of each other. It can sometime lead to inaccurate predictions as in real-world scenario, the features may be correlated. It can be used in various tasks like document classification, email filtering, sentiment analysis, medical diagnosis, real-time predictions and many more. Some of the Naïve Bayes Classification models are Multinomial NB, Bernoulli NB and Gaussian NB. The Gaussian NB is used for continuous data and it assumes that the features follow a normal distribution. The Multinomial NB is used for discrete data where the data can be represented as word counts or frequencies. The Bernoulli NB is

used for binary or boolean data. Additionally, there are other hybrid models that can outperform these standard types.

In this study, we aim to explore and enhance the performance of Naïve Bayes Classifiers on the Estimation of Obesity Levels Based on Eating Habits and Physical Condition dataset, which provides a rich set of features related to individuals' dietary habits and physical attributes. This dataset is particularly suitable for our study as it includes both categorical and continuous variables, offering a comprehensive testing ground for the enhanced Naive Bayes model.

II. Literature review

The paper [1] employed Naïve Bayes Classifier to predict lecturer satisfaction levels based on these attributes. The results demonstrated that the Naive Bayes method achieved an accuracy of 85.48%,

with a precision of 81.08% and a recall of 93.75%. It concludes that the Naive Bayes method can be a valuable tool for predicting and improving lecturer satisfaction and, consequently, institutional performance.

The study reported in the paper [2] carried out a comparative examination of three Naïve Bayes models: Gaussian, Multinomial, and Bernoulli in the context of health-related classification tasks. The Multinomial Naïve Bayes model achieved the highest mean accuracy of 97.07%, outperforming the Gaussian and Bernoulli models on the heart disease and breast cancer dataset. This shows that the Multinomial model's capacity to handle discrete data well makes it especially useful for classification problems involving health-related data.

The paper [3] compares Naive Bayes and Random Forest classifiers to predict clinical diseases such as diabetes, coronary heart disease, and breast cancer. In order to increase the models' accuracy, the study underlines how crucial it is to handle missing values and look for connections while preprocessing data. In comparison to the Naive Bayes classifier, the Random Forest classifier performed better, especially when processing high-dimensional data and recognizing key features. The Random Forest model outperformed the Naive Bayes model with much higher accuracy rates: 98.88% for diabetes, 97.96% for heart disease, and 99.50% for breast cancer.

This paper [4] incorporates weighting and Laplace calibration, the enhanced algorithm addresses the limitations of the traditional Naive Bayes classifier, particularly its assumption of attribute independence and vulnerability to sparse data issues. This improved algorithm demonstrated high accuracy, achieving over 99% accuracy with large sample sizes and maintaining above 95% accuracy with fewer attributes. When applied to traffic violation data, the improved algorithm significantly outperformed the original Naive Bayes classifier and multiple logistic regression models, achieving an overall correct classification

rate of 92%, compared to 49.5% for the original Naive Bayes classifier.

III. METHODOLOGY

A. Theoretical formulation

Naive Bayes classification is a supervised learning algorithm that uses the Naive Bayes Theorem to make predictions about the class labels of new data points based on their feature values. It is called "naive" because it makes the simplifying assumption that all features are conditionally independent given the class label. This means that the presence or absence of one feature does not affect the presence or absence of another feature, given the class label.

B. Mathematical Formulation

Standard Naive Bayes

Bayes' Theorem provides a way to update the probability estimate for a hypothesis as more evidence or information becomes available. In the context of classification, Bayes' Theorem can be used to calculate the probability of a class given the features of an instance.

The theorem is stated as:

$$P(c | x) = \frac{P(x | c) \cdot P(c)}{P(x)}$$

$P(c | x)$ is the posterior probability of class

$P(c | x)$ is the likelihood of the feature vector

$P(c)$ is the prior probability of class

$P(x)$ is the evidence

Gaussian Naïve Bayes Classifier

It assumes that the continuous values associated with each class are distributed according to a Gaussian (normal) distribution. This is suitable for datasets where the features are continuous and normally distributed.

For a feature vector $x = (x_1, x_2, \dots, x_n)$, the likelihood $P(x | c)$ is computed using the Gaussian distribution:

$$P(x_i | c) = \frac{1}{\sqrt{2\pi\sigma_c^2}} \exp\left(-\frac{(x_i - \mu_c)^2}{2\sigma_c^2}\right)$$

Where μ_c and σ_c^2 are the mean and standard deviation of the feature x_i for class c .

Multinomial Naïve Bayes Classifier

It is typically used for discrete data, particularly useful in text classification tasks where data can be represented as word counts or term frequencies.

The distribution of word vector count is parametrized by vectors $\theta_c = (\theta_{c_1}, \theta_{c_2}, \dots, \theta_{c_n})$ for each class c , where n is the number of features (in text classification, the size of the vocabulary) and θ_{c_i} is the probability $P(x_i | c)$ of feature i appearing in a sample belonging to class c . The parameter θ_c is estimated by a smoothed version of maximum likelihood, i.e. relative frequency counting:

$$\theta_{c_i} = \frac{N_{yi} + \alpha}{N_y + \alpha n}$$

Where N_{yi} is the number of times feature i appear in a sample of class c in the training set

N_y is the total count of all features for class c

$\alpha \geq 0$ is the smoothing factor that prevents zero probabilities in further computations. Setting $\alpha = 1$ is called Laplace smoothing, while $\alpha < 1$ is called Lidstone smoothing [5]

Bernoulli Naïve Bayes Classifier

It is suitable for binary/boolean features, where each feature represents the presence or absence of a characteristic.

The decision rule for Bernoulli Naïve Bayes Classifier is based on:

$$P(x_i | c) = P(i | c)x_i + (1 - (i | c))(1 - x_i)$$

On some datasets, particularly those with shorter texts, Bernoulli Naïve Bayes may perform better.

Hybrid model

In hybrid models, the combined probability is computed by integrating the probabilities from

Gaussian and Categorical Naive Bayes classifiers. The system block diagram is shown in Figure 1 For instance, if we have continuous features x_G and categorical features x_C :

$$P(c | x) = P(c | x_G) \cdot P(c | x_C)$$

Where $P(c | x_G)$ is computed using the Gaussian Naive Bayes formula and $P(c | x_C)$ using the Multinomial or Bernoulli Naive Bayes formula, depending on the nature of the categorical features.

Correlation matrix

The correlation matrix is a valuable tool in the preprocessing and analysis stages of developing a Naive Bayes classifier, despite the classifier's fundamental assumption of feature independence. The values range from -1 to 1, where values closer to 1 or -1 indicate strong positive or negative correlations, respectively, and values near 0 indicate weak or no correlation. Although Naive Bayes assumes feature independence, understanding feature dependencies is crucial. The correlation matrix helps in identifying these dependencies, allowing you to consider alternative models or hybrid approaches. Features with low correlation to the target variable might contribute less to the prediction accuracy. The correlation matrix helps in feature selection by highlighting which features have stronger relationships with the target variable, thus allowing you to focus on the most relevant features and discard less informative ones.

C. Evaluation

In classification and predictive modeling, accuracy and precision are fundamental metrics used to evaluate the performance of a model.

Accuracy

Accuracy is a measure of the overall correctness of a model. It represents the proportion of correct predictions (both true positives and true negatives) out of all predictions made by the model.

Mathematically, accuracy is defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

TP (True Positives) are instances where the model correctly predicts the positive class.

TN (True Negatives) are instances where the model correctly predicts the negative class.

FP (False Positives) are instances where the model incorrectly predicts the positive class.

FN (False Negatives) are instances where the model incorrectly predicts the negative class.

If there is no class imbalance problem, a high accuracy indicates that the model is making a large number of correct predictions relative to the total number of predictions.

Precision

Precision is a measure of the correctness of positive predictions made by the model. It represents the proportion of true positive predictions out of all positive predictions (true positives and false positives).

Mathematically, precision is defined as:

$$Precision = \frac{TP}{TP + FP}$$

Recall

Recall is a measure of the model's ability to correctly identify positive instances. It represents the proportion of true positive predictions out of all actual positive instances.

$$Recall = \frac{TP}{TP + FN}$$

F1-score

The F1-score is the harmonic mean of precision and recall. It provides a single metric that balances both the precision and recall of the model.

$$F1 - score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

D. Instrumentation tools

Python was used throughout, making use of a number of tools and modules to speed up the process of training the model and analyzing data.

Our main platform for coding and experimenting was Jupyter Notebook. The pandas package was utilized to load and handle the dataset, offering effective data structures and capabilities for data analysis. We used NumPy, which provides extensive mathematical functions, for numerical calculations. We utilized matplotlib and Seaborn with powerful plotting features, to create visually appealing and educational representations of the data and findings. The Naive Bayes classifiers were trained using the scikit-learn (sklearn) library, which makes use of a large array of machine learning tools and techniques. To guarantee appropriate model evaluation and validation, the dataset was divided into training and testing sets using the scikit-learn `train_test_split` function. All of these technologies worked together to make our analytical and modeling jobs more thorough and effective.

E. Data Preprocessing Pipeline

Data preprocessing is a crucial step in any machine learning project as it prepares raw data for further analysis and modeling. This process involves several steps to ensure the data is clean, consistent, and suitable for training machine learning algorithms. To ensure a systematic and reproducible preprocessing workflow, we implemented a data preprocessing pipeline using Python libraries such as pandas and scikit-learn. The pipeline includes steps for handling missing values, normalizing and standardizing features, encoding categorical variables, selecting relevant features, and splitting the dataset into training and testing sets. Here, we discuss the key steps involved in data preprocessing for our study.

Feature Selection

Feature selection involves identifying and selecting the most relevant features for the model. We can use the correlation matrix for this purpose. This can improve model performance by reducing overfitting and computational cost. It is important to select relevant features based on their importance or correlation with the target variable.

Handling Missing Values

Missing data can introduce biases and reduce the accuracy of the model. There are several techniques to handle missing values:

Imputation: It involves replacing the missing values with a specific value such as the mean, median, or mode of the column. For categorical features, we can replace the missing value with the most frequent category.

Removal: In some cases, it may be appropriate to remove rows or columns with a significant number of missing values, particularly if the proportion of missing data is high.

Normalization

Normalization and standardization are techniques used to scale numerical features to a standard range, typically to ensure that no single feature dominates the model due to its scale. We can scale the data to a range of $[0, 1]$ or $[-1, 1]$ when the features have different ranges. We can also transform the data to have a mean of zero and a standard deviation of one when the data follows a normal distribution.

Encoding Categorical Variables

Gaussian Naïve Bayes Classifier require numerical input, so categorical variables need to be converted into a numerical format. We need to convert categorical variables into numeric form using one-hot encoding or label encoding.

Outlier Detection and Removal

Outliers can skew the results and adversely affect the model's performance. We can use Z-score, IQR (Interquartile Range) method, or visual inspection through boxplots.

Splitting the Dataset

It is essential to divide the dataset into training and testing sets, commonly using a ratio such as 70:30 or 80:20. The training set is used to train the model, while the testing set is used to evaluate its performance.

F. Data Analysis and Visualization

Plots of various kinds are used to illustrate the insights obtained from the dataset and model training. The dataset's features and their instances are displayed as separate histogram plots for every feature. Horizontal bar plots are another way that the training data's outcomes are displayed. A heatmap is used to show how various attributes are correlated. A confusion matrix is used to visualize the model's performance and outcomes after it has been trained.

IV. RESULT

The results of our comparison between hybrid models and Gaussian Naive Bayes for improved classification using the "Estimation of Obesity Levels Based on Eating Habits and Physical Condition" dataset are shown in this section. To find the best method for predicting obesity levels, we tested the performance of a variety of Naive Bayes classifiers, such as Gaussian Naive Bayes, Categorical Naive Bayes, and hybrid models that incorporate both types.

A. Dataset Interpretation

The dataset "Estimation of Obesity Levels Based on Eating Habits and Physical Condition" contains information from individuals in Mexico, Peru and Colombia. It has 2111 instances and 17 features out of which 9 are categorical, 8 are numeric which are presented in Table 1. This dataset is multivariate and suitable for classification, regression, and clustering tasks.

The histograms show that the distribution of Age is skewed towards younger individuals, with a peak around the early 20s. There is a noticeable drop-off in frequency as age increases, indicating that the dataset primarily consists of younger individuals between 15 and 35. The height is roughly normally distributed. The Weight distribution is multimodal, with several peaks. There is a central peak around 70-80 kg, but additional smaller peaks indicate the presence of distinct groups within the data. For FAF and CH2O, the data is not evenly distributed, with

more individuals reporting specific frequencies of physical activity.

B. FITTING THE GAUSSIAN NAÏVE-BAYES CLASSIFIER

The Gaussian Naïve Bayes Classifier was fit to the training set. We used 80% for train and 20% for test. While fitting the Gaussian Naïve Bayes Classifier, we observed that it achieved an accuracy of 64%. It performed very well in classifying Obesity Type III with an F1-score of 1.00 as seen in Classification report shown in Table 2. All the 63 instances were correctly classified as Obesity Type III with no misclassifications. However, it struggled with Normal Weight and Overweight Levels I and II. The confusion matrix reveals that the model often confuses Normal Weight with Insufficient Weight and Overweight categories, as well as Overweight Level I and II with other obesity types. This implies that although the model performs well in detecting extreme obesity, it has trouble differentiating between weight categories that are more in the middle. It may take more feature engineering and refining to increase the model's accuracy in classifying these unclear circumstances.

C. FITTING THE CATEGORICAL NAÏVE BAYES MODEL

Here, we present the results of fitting the Categorical Naïve Bayes model to the "Estimation of Obesity Levels Based on Eating Habits and Physical Condition" dataset. We evaluated the model using two binning techniques: equal width binning and equal frequency binning.

Equal Width Binning

The model achieved an overall accuracy of 70% which is slightly better than Gaussian Naïve Bayes Classifier. It performed exceptionally well in classifying Obesity Type III, with precision, recall, and F1-score all close to 1.00. The model showed moderate performance for Insufficient Weight, Normal Weight, and Obesity Type II, with F1-scores ranging from 0.62 to 0.86 that can be seen in Table 3. The performance was lower for

Overweight Level I and II, with F1-scores of 0.45 and 0.42, respectively.

Equal Frequency Binning

The result of equal frequency binning is not very different from the result of equal width binning. The model achieved an overall accuracy of 71%, slightly higher than the equal width binning approach. It performed well in classifying Obesity Type III, similar to the equal width binning. The model showed improved performance for Insufficient Weight and Normal Weight, with F1-scores of 0.81 and 0.72, respectively as seen in Table 4. The performance for Overweight Level I and II remained lower but showed a slight improvement, with F1-scores of 0.53 and 0.44, respectively.

However, both approaches struggled with Overweight Level I and II, indicating the need for further refinement and different modeling techniques to improve classification accuracy for these categories.

G. FITTING A HYBRID MODEL

The hybrid model is a combination of Gaussian Naïve Bayes for continuous data and Categorical Naïve Bayes for categorical data. It shows an exceptional performance across all the classes with a few misclassifications. This hybrid model achieved an accuracy of 99% on the dataset as shown in Table 5 indicating that 99% of total predications made by the model are correct. The off-diagonal elements, representing the misclassifications are nearly zero in the confusion matrix in Figure 10 Confusion Matrix of Hybrid Model Figure 10. The hybrid model outperformed individual Gaussian and Categorical Naïve Bayes models by handling the mixed data types.

V. DISCUSSION AND ANALYSIS

We first visualized the target classes to see if there is class imbalance. The bar plot shown in Figure 2 shows that the class distribution is nearly uniform. Hence since there is no class imbalance problem, we can use accuracy as a metric to evaluate the performance of the classifiers.

We can see the histogram plot of all the features in Figure 5 whereas Figure 6 shows the histogram plot of continuous features. We need histogram plots to tackle the continuous variables. By analyzing the histogram plots, we should fit a known approximate distribution to the continuous features. Here, we fit normal distribution to age, height and weight as these features mimic the normal distribution.

From the correlation matrix in Figure 4 we can conclude that there is a moderate positive correlation between height and gender and height and weight, suggesting that height varies noticeably with gender and that taller individuals tend to weigh more. There is a moderate positive correlation between weight and having a family history of overweight, implying that individuals with a family history of overweight are likely to weigh more. Most other features exhibit weak correlations, suggesting limited linear relationships among them. We can also observe that alcohol consumption is more correlated with obesity than smoking. These correlations highlight the importance of certain features such as height, weight, and family history in the context of obesity.

The Gaussian Naïve Bayes model struggled with categorical data, while the Categorical Naïve Bayes model had limitations with continuous features. By combining these models, we leveraged their strengths, resulting in superior classification performance.

We tried hybrid models like the hybrid of Gaussian Naïve Bayes model and Multinomial Naïve Bayes model but this hybrid achieved an accuracy of only 63% for this dataset. We used 80:20 split for training and testing i.e. 1688 for train, 423 for test.

The hybrid of Gaussian and Categorical Naïve Bayes model outperformed the separate models by a large margin, achieving an excellent overall accuracy of 99% through the combination of Gaussian and Categorical methods. The strength of this model is its capacity to process continuous and categorical variables in an efficient manner, which

results in good recall, precision, and F1-scores for all obesity categories. Despite its high accuracy, the model did exhibit minor misclassifications in categories like Obesity Type I and III. These could be due to overlapping feature distributions or inherent noise in the data. The algorithm performed exceptionally well in assigning almost perfect scores to high obesity levels (such as Obesity Type III). The accuracy of the different Naïve Bayes classifiers have been shown in Figure 12

The AUC of 0.99 in ROC curve in Figure 11 suggests that the hybrid Naïve Bayes model is highly effective with minimal overfitting and high consistency. The curve's proximity to the top-left corner of the plot further illustrates the model's robustness, as it shows that the model achieves high sensitivity without compromising on specificity. The narrow spread of the standard deviation band around the mean ROC curve indicates that the model's performance is consistent across different cross-validation folds, which adds to the reliability of the results.

We discovered via numerical simulation that the accuracy rate of the discrimination analysis of the naive Bayesian classification algorithm fluctuates significantly when the sample size is small. However, as the sample size increases, the fluctuation gradually gets smaller and the overall trend tends to be stable, with the accuracy reaching more than 99%.

VI. CONCLUSION

This study successfully applied Naïve Bayes classifiers to predict obesity levels based on eating habits and physical conditions. The preprocessing steps, including handling missing data, label encoding, and discretization of features, were crucial in preparing the dataset for analysis. The hybrid Naïve Bayes model, which combined Gaussian and Categorical Naïve Bayes classifiers, showed better performance compared to individual models, achieving an accuracy of 63%. Additionally, addressing class imbalance and incorporating more data could improve the robustness and accuracy of the predictions.

Naive Bayes classifiers, while simple and effective in many applications, have notable drawbacks that can limit their performance. The term "Naive" in Naive Bayes stems from its fundamental assumption of feature independence. This assumption is considered "naive" because it simplifies the complex reality where features are often interdependent. By assuming independence, the classifier simplifies the computation of probabilities, making the model straightforward and computationally efficient. However, this naivety also leads to the limitations discussed, as it does not capture the interactions between features that could be crucial for accurate predictions. The primary issue is the assumption of feature independence, meaning it assumes that all features contribute to the outcome independently of each other. This assumption rarely holds true in real-world data, where features often exhibit correlations. When such dependencies exist, the Naive Bayes classifier may produce suboptimal results.

Additionally, Naive Bayes can struggle with small training datasets, leading to overfitting or underfitting. It also tends to assign zero probability to unseen feature-label combinations in the training data, which can be problematic for classification. Despite these drawbacks, the simplicity and efficiency of Naive Bayes often make it a useful baseline model.

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KRISTINA BHANDARI is a computer engineering undergraduate student at IOE, Thapathali Campus. She is eager to explore the latest advancements in AI and aims

to make significant contributions to the field. (THA077BCT022)



NAMITA BHATTA is a computer engineering undergraduate currently in her fourth year at IOE, Thapathali Campus. Her goal is to leverage her knowledge and expertise to

make a significant impact in the field. (THA077BCT048)

APPENDIX

TABLES

Table 1 Dataset Description

Feature	Description	Type
<i>Gender</i>	Gender of the individual (Male/Female)	Categorical
<i>Age</i>	Age of the individual in years	Continuous
<i>Height</i>	Height of the individual in meters	Continuous
<i>Weight</i>	Weight of the individual in kilograms	Continuous
<i>family_history_with_overweight</i>	Family history of being overweight (yes/no)	Categorical
<i>FAVC</i>	Frequent consumption of high caloric food (yes/no)	Categorical
<i>FCVC</i>	Frequency of consumption of vegetables (0-3)	Continuous
<i>NCP</i>	Number of main meals per day (1-5)	Continuous
<i>CAEC</i>	Consumption of food between meals (no/Sometimes/Frequently/Always)	Categorical
<i>SMOKE</i>	Smoking habits (yes/no)	Categorical
<i>CH2O</i>	Daily consumption of water (in liters) (1-3)	Continuous
<i>SCC</i>	Calories consumption monitoring (yes/no)	Categorical
<i>FAF</i>	Physical activity frequency per week (0-3)	Continuous
<i>TUE</i>	Time using technology devices per day (0-2)	Continuous
<i>CALC</i>	Consumption of alcohol (no/Sometimes/Frequently/Always)	Categorical
<i>MTRANS</i>	Transportation used (Automobile/Motorbike/Bike/Public_Transportation/Walking)	Categorical
<i>NObesidad</i>	Obesity level (Insufficient_Weight/Normal_Weight/Overweight_Level_I/Overweight_Level_II/Obesity_Type_I/Obesity_Type_II/Obesity_Type_III)	Categorical

Table 2 Classification report of Gaussian Naïve Bayes Classifier

Class	Precision	Recall	F1-score	Support
Insufficient_Weight	0.70	0.91	0.79	56
Normal_Weight	0.68	0.34	0.45	62
Obesity_Type_I	0.44	0.64	0.52	78
Obesity_Type_II	0.66	0.91	0.77	58
Obesity_Type_III	1.00	1.00	1.00	63
Overweight_Level_I	0.49	0.30	0.37	56
Overweight_Level_II	0.57	0.32	0.41	56
accuracy			0.64	423
macro avg	0.65	0.63	0.62	423
weighted avg	0.64	0.64	0.62	423

Table 3 Classification report of Categorical Naïve Bayes Classifier for equal width binning

Class	Precision	Recall	F1-score	Support
Insufficient_Weight	0.72	0.82	0.77	56
Normal_Weight	0.62	0.61	0.62	62
Obesity_Type_I	0.77	0.60	0.68	78
Obesity_Type_II	0.81	0.93	0.86	58
Obesity_Type_III	0.98	1.00	0.99	63
Overweight_Level_I	0.68	0.41	0.45	56
Overweight_Level_II	0.38	0.46	0.42	50
accuracy			0.70	423
macro avg	0.68	0.69	0.68	423
weighted avg	0.70	0.70	0.69	423

Table 4 Classification report of Categorical Naïve Bayes Classifier for equal frequency binning

Class	Precision	Recall	F1-score	Support
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Insufficient_Weight	0.85	0.79	0.81	56
Normal_Weight	0.64	0.82	0.72	62
Obesity_Type_I	0.75	0.58	0.65	78
Obesity_Type_II	0.75	0.83	0.79	58
Obesity_Type_III	0.98	1.00	0.99	63
Overweight_Level_I	0.55	0.52	0.53	56
Overweight_Level_II	0.44	0.44	0.44	50
accuracy			0.71	423
macro avg	0.71	0.71	0.71	423
weighted avg	0.72	0.71	0.71	423

Table 5 Hybrid Model(Gaussian and Categorical)

Class	Precision	Recall	F1-score	Support
Insufficient_Weight	1.00	1.00	1.00	56
Normal_Weight	1.00	1.00	1.00	62
Obesity_Type_I	1.00	0.96	0.98	78
Obesity_Type_II	1.00	1.00	1.00	58
Obesity_Type_III	0.94	1.00	0.97	63
Overweight_Level_I	1.00	1.00	1.00	56
Overweight_Level_II	1.00	0.98	0.99	50
accuracy			0.99	423
macro avg	0.99	0.99	0.99	423
weighted avg	0.99	0.99	0.99	423

FIGURES

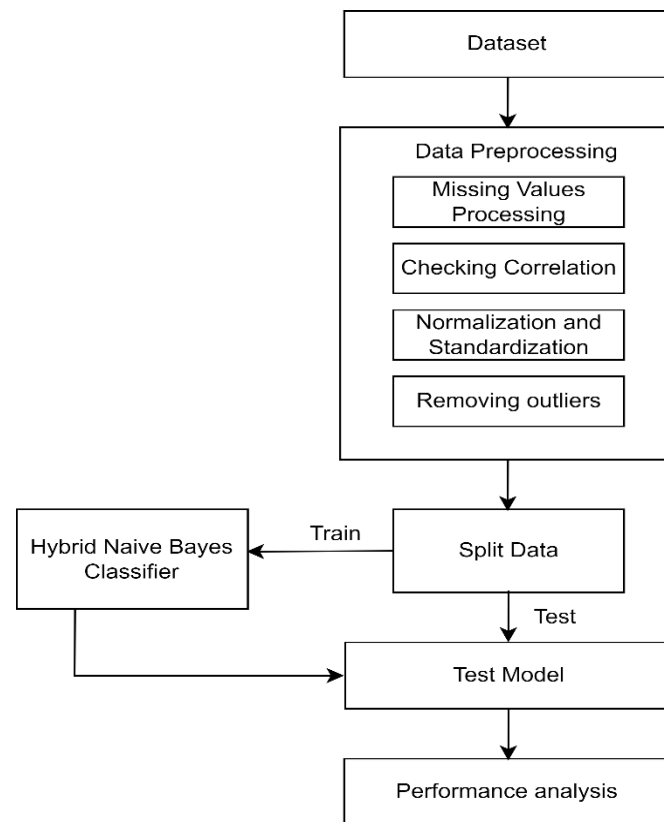


Figure 1 System Blockdiagram

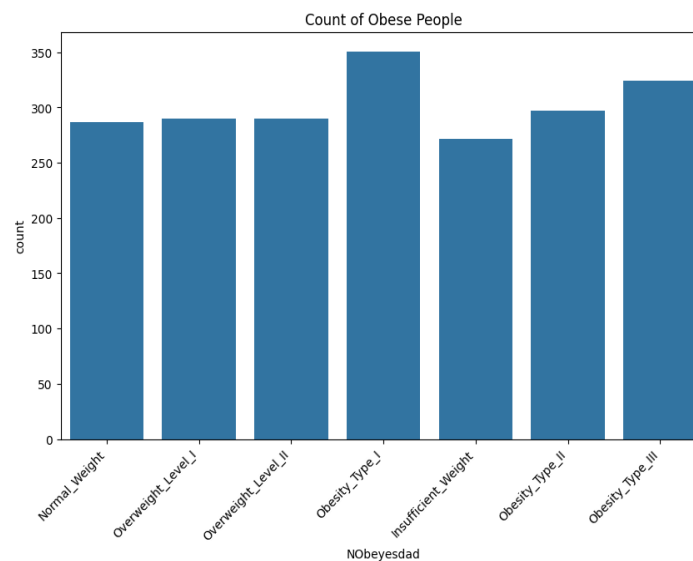


Figure 2 Class Distribution of the Dataset

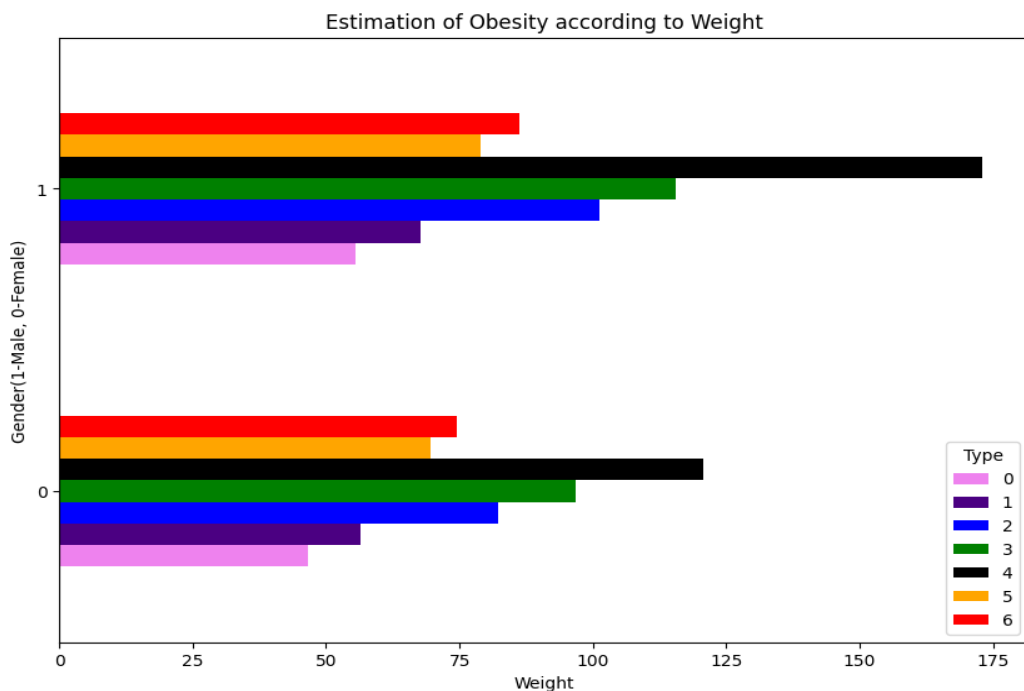


Figure 3 Class Distribution according to weight

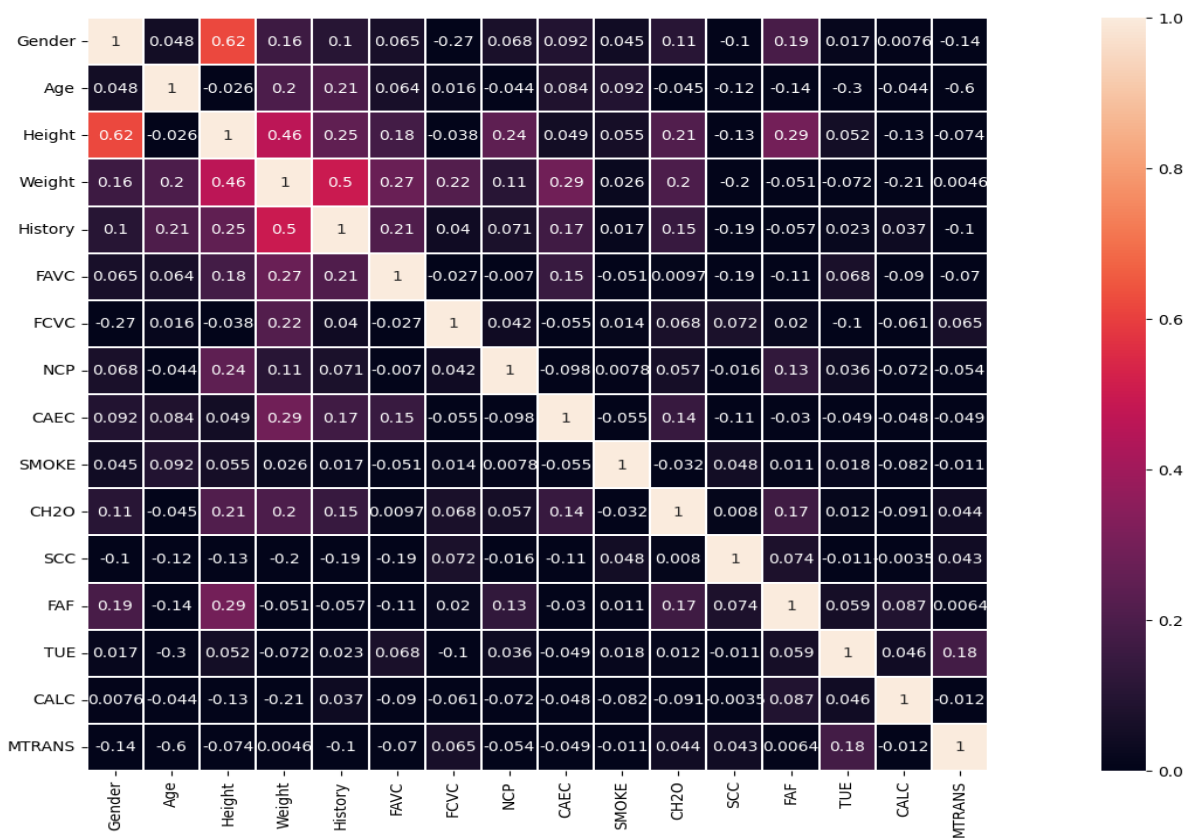


Figure 4 Correlation (Pearson) between features

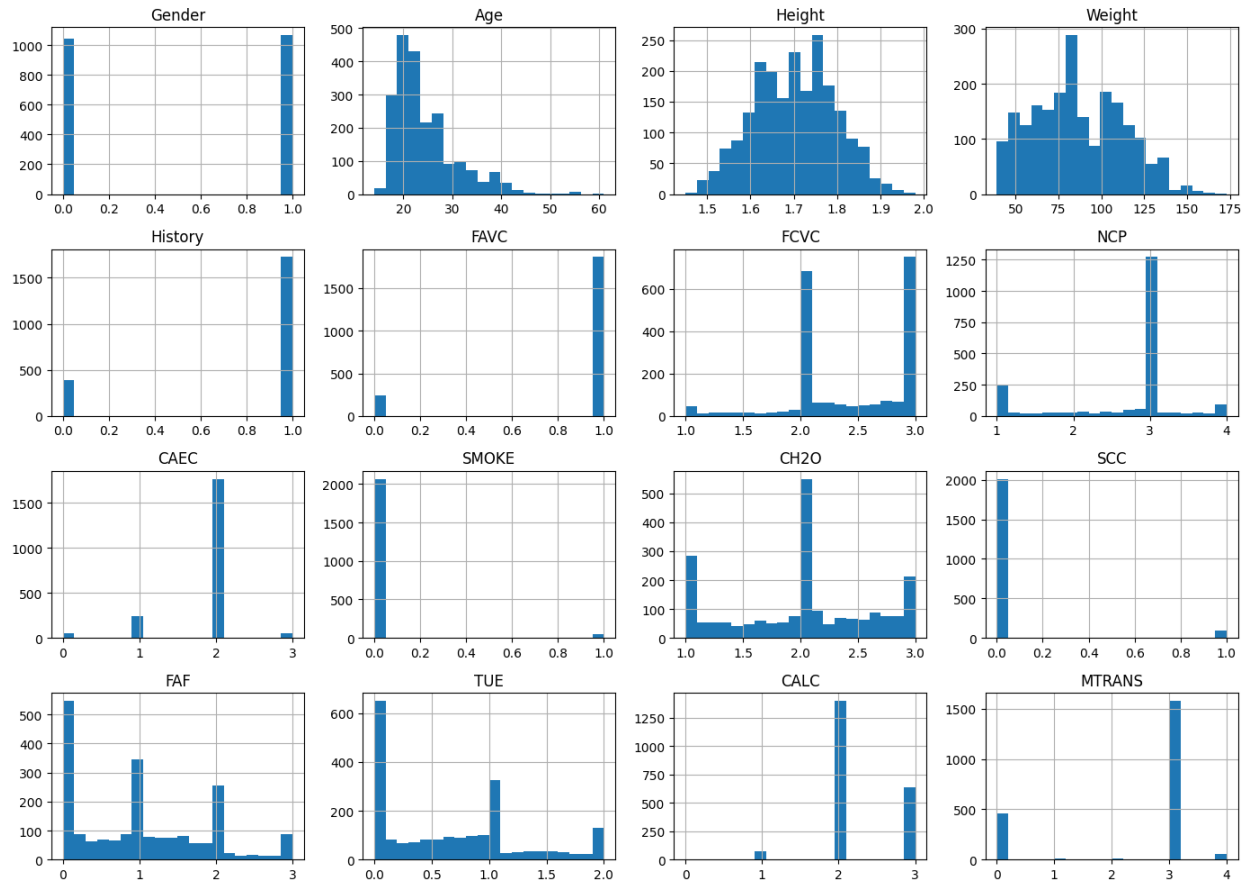


Figure 5 Histogram plot of all features

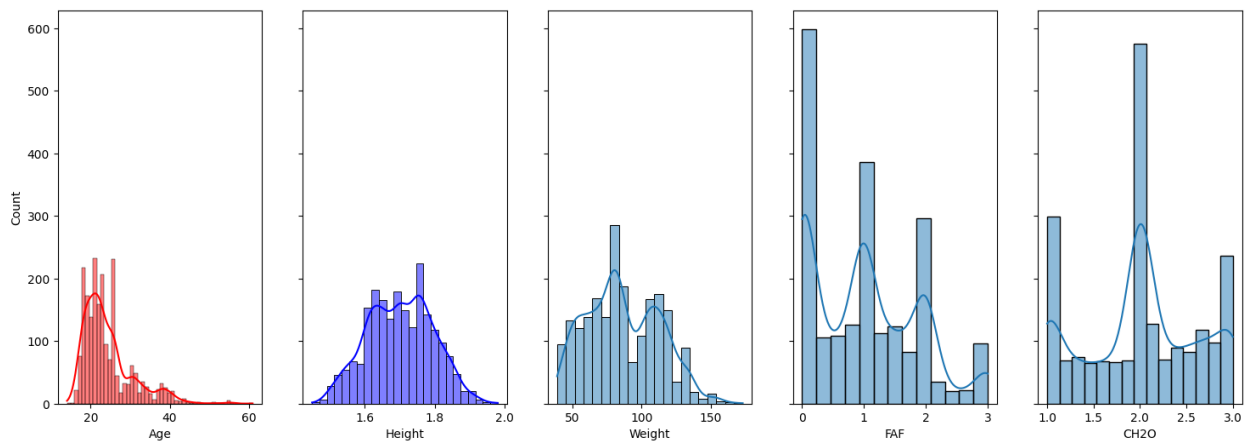


Figure 6 Histogram plot of Continuous Feature

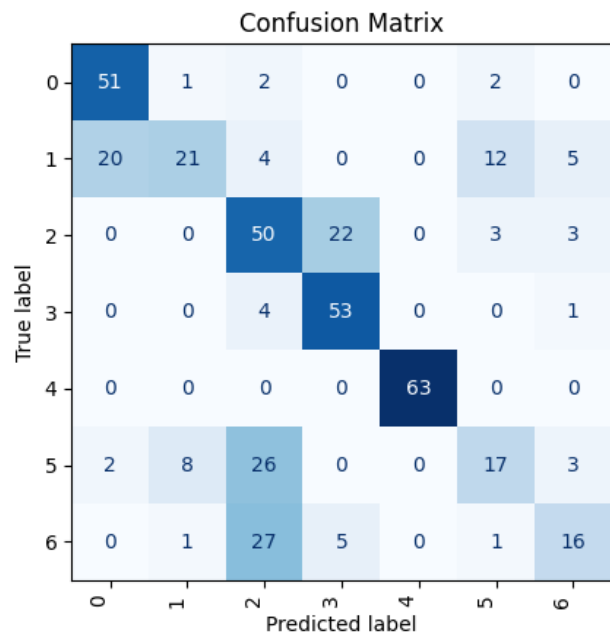


Figure 7 Confusion matrix Gaussian Naïve Classifier

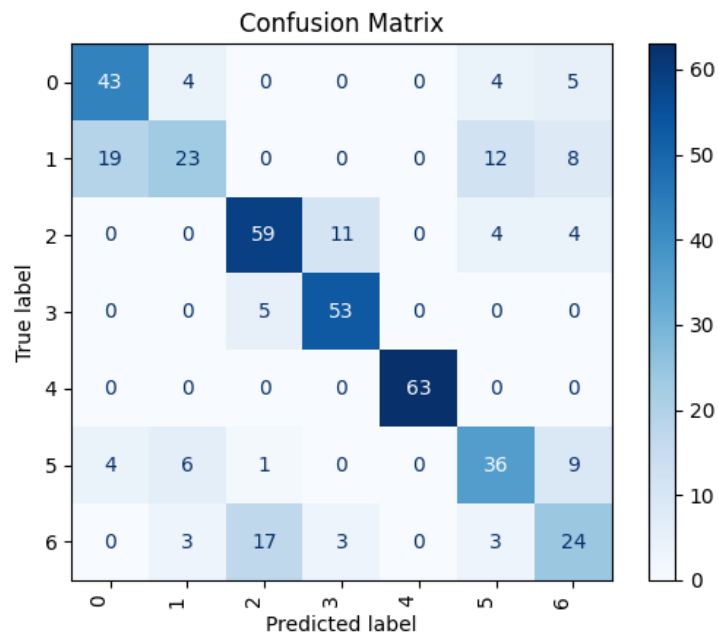


Figure 9 Confusion matrix Categorical Naïve Classifier for equal frequency binning

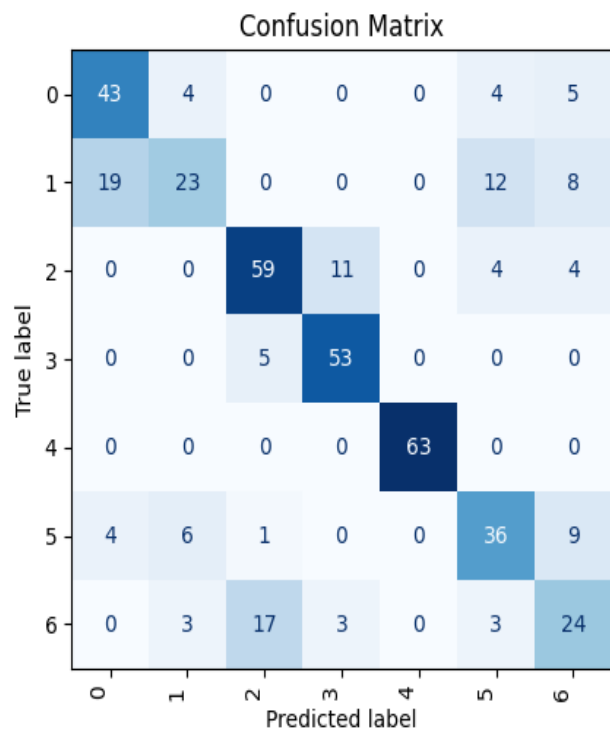


Figure 8 Confusion matrix Categorical Naïve Classifier for equal width binning

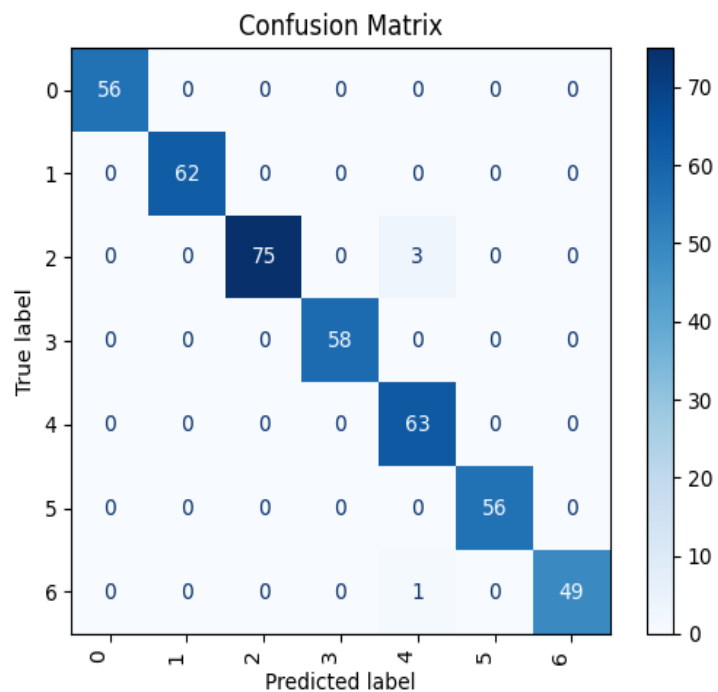


Figure 10 Confusion Matrix of Hybrid Model

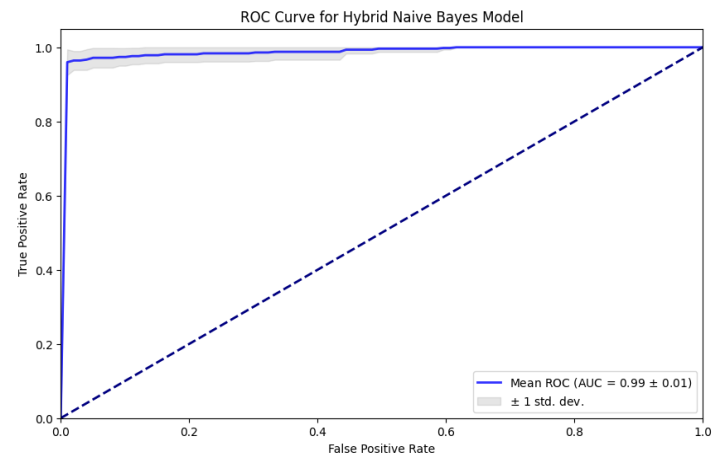


Figure 11 ROC curve of Hybrid Naïve Bayes Model

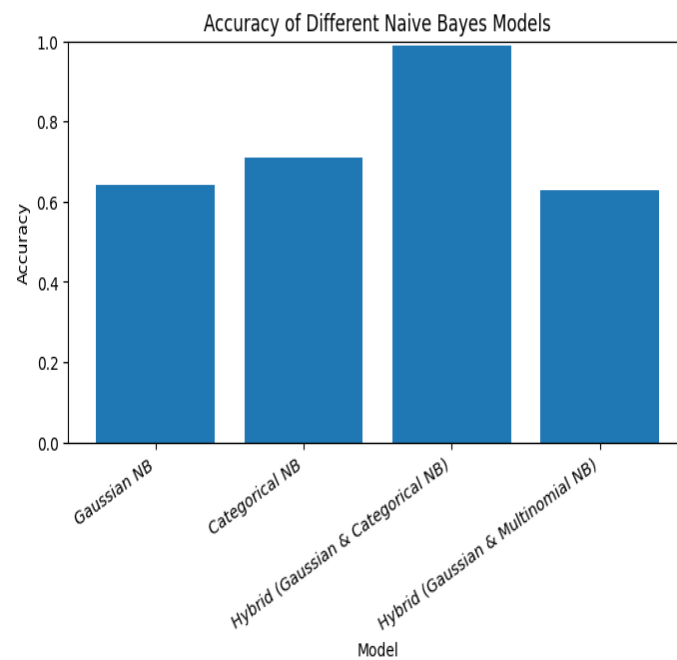


Figure 12 Comparison of Different Naïve Bayes Classifier